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SDD-PINN: Physics-informed neural network for single droplet drying

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ABSTRACT

Single droplet drying, a fundamental process in spray drying, presents a challenging nonlinear moving boundary diffusion problem. This process is described by a parabolic partial differential equation in a shrinking spherical domain with a Robin mass-transfer boundary condition. Despite extensive recent applications of physics-informed neural networks (PINNs) to solve PDEs, vanilla PINNs often struggle with time-dependent, moving boundary transport problems. This study develops SDD-PINN, a compact and reproducible PINN framework that remains reliable on such evolving domains. To decouple the training from domain shrinkage, we first map the problem to a fixed unit domain via a radius transformation, $\xi = r/a(t)$, yielding a nonlinear advection-diffusion PDE that preserves mass conservation (baseline PINN). Building on this, a compact, physics-motivated PINN recipe is presented comprising: (i) hard enforcement of the initial condition, (ii) a squared transformed radius (ξ^2) as a symmetry-consistent coordinate input to promote regularity near the spherical center, (iii) a logarithmic time reparameterization θ from non-dimensional time τ , (iv) smart collocation sampling, and (v) a combined Adam + L-BFGS optimization schedule. A 2^5 full-factorial experimental design (soft vs. hard initial condition, ξ vs. ξ^2 , τ vs. θ , uniform vs. smart sampling, Adam vs. Adam + L-BFGS) is first conducted on a reference regime to identify a progressive enhancement path. The selected configuration, SDD-PINN (hard initial condition, ξ^2 feature, and θ as input, uniform sampling, and a combined Adam + L-BFGS optimizer), resulted in a mean relative L_2 error 0.021 ± 0.010 against a traditional Crank–Nicolson (CN) reference. Validation across multiple drying conditions spanning shrinkage intensity, diffusivity, and particle size with multi-seed repeats showed that relative to the CN reference, the unified recipe yields a mean relative L_2 error ranging from 2.69×10^{-3} to 3.30×10^{-2} . These results provide a reproducible, PINN recipe for single droplet drying.

1. Introduction

Spray drying is a critical industrial process for producing powders in the food, pharmaceutical, and chemical industries (Samborska et al., 2023; Tran et al., 2017). Because of the complex interactions among various physical phenomena in a spray dryer, single droplet drying (SDD), where an isolated droplet is studied under controlled conditions mimicking spray drying, is an essential step for understanding the temperature and moisture histories, droplet shrinkage, and morphological evolution (Tran et al., 2016). Single droplet drying models are commonly categorized into spatially concentrated (global) and spatially distributed (resolving) models. In spatially concentrated models, the intra-droplet profiles are neglected, and only droplet-averaged quantities evolve, whereas in spatially distributed models, these profiles are explicitly computed.

As the solvent is removed, steep internal solids or solute

concentration gradients develop; the near-surface solute concentration governs the interfacial flux, triggers crust formation (“locking point”), and controls the transition to the second drying stage. Predicting these events requires profile-level access to the concentration field, not just bulk averages, along with reliable surface flux histories and mass balance checks. Therefore, spatially distributed models are essential because they can represent crust formation and the transition to the second drying stage. The concentration gradients can also help explain whether the morphology evolves toward a hollow shell or a dense particle. Spatially resolving models formulate drying and structure formation as a Fickian diffusion partial differential equation (PDE) in the shrinking droplet with a Robin-type boundary condition (BC) at the surface and are more physically faithful. Therefore, they provide greater insights into the transferability from SDD to spray tower conditions, albeit at a higher computational cost (Tran, Jaskulski, Avila-Acevedo, et al., 2017).

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Classical numerical solvers (e.g., finite-volume/finite-difference schemes) for spatially distributed models are accurate and well-established, but typically require meshing, careful handling of moving interfaces, and repeated runs for each operating point when scanning conditions or performing sensitivity studies. They are also challenged by inverse problems. Physics-informed neural networks (PINNs) have become a prominent framework for solving forward and inverse problems governed by PDEs. By embedding physical laws (PDEs and their initial/boundary conditions (IC/BCs)) into the neural networks (Karniadakis et al., 2021), PINNs learn a smooth, differentiable surrogate of the spatiotemporal field without labeled data and, once trained, can act as fast mesh-free surrogate models that enable rapid predictions at low computational costs (Manavi et al., 2025). Recent studies have applied PINNs to a range of heat and mass transfer problems (Jalili et al., 2024; Manavi et al., 2024, 2025; M. Wang et al., 2025; Zhao et al., 2024).

Despite many successful applications, a substantial body of evidence documents the characteristic failure modes of “vanilla” PINNs, particularly for time-dependent transport and multiscale regimes (Daw et al., 2022; Gao et al., 2023; Krishnapriyan et al., 2021; S. Wang, Teng, et al., 2021; Wang, Wang, et al., 2021; Zhang et al., 2026). Remedies explored in the literature include loss function design and imbalance analyses (Frerichs-Mihov et al., 2024), reformulations via coordinate/characteristics-based transformations (Y. Chen et al., 2025), methodologies for coupled moving-boundary PDEs (Kathane and Karagadde, 2024), alternative learners for moving-boundary (Stefan) problems (Ren et al., 2025), and domain-specific critical evaluation of PINNs (H. Chen et al., 2022), each aiming to mitigate these shortcomings.

Loss imbalance issues arise when the gradients contributed by different components of the physics loss (PDE residual vs. IC/BC terms) differ by orders of magnitude, causing the optimizer to prioritize the easiest constraints while neglecting the hardest ones (Krishnapriyan et al., 2021; S. Wang, Teng, et al., 2021). Spectral bias refers to the tendency of standard neural networks to prioritize learning low-frequency components first, while facing difficulties with sharp spatiotemporal features that emerge during time evolution. In the context of droplet drying, such challenges correspond to the formation of steep internal concentration gradients near the surface, which critically affect the interfacial mass flux and crust formation. When solving PDEs, this can trap the training in trivial or excessively smooth solutions (Wang, Wang, et al., 2021; Xu et al., 2025). In time-dependent problems, where training is performed over the entire space-time domain simultaneously, optimization may violate temporal causality, resulting in “propagation failures”. In such cases, PINNs are unable to carry the information from the IC forward to later times (Daw et al., 2022).

Moving boundaries add significant complexity to PINN training. In drying, dissolution, solidification, and related processes, the spatial domain evolves over time, and the interface conditions couple the interior field to the boundary motion. Prior work has shown that vanilla PINNs often struggle to solve such problems because of the difficulty in enforcing interface constraints and nonlinear interactions among loss components; geometric terms in spherical coordinates can further amplify the local stiffness of the numerical model (Kathane and Karagadde, 2024; Ren et al., 2025). The drying of spherical particles is particularly sensitive: the regularity at the center must be preserved whereas the surface typically adopts a Robin BC to represent external mass transfer. Both factors can destabilize training if not handled carefully.

The SDD is a canonical benchmark for diffusion in spherical coordinates on a shrinking domain with a Robin BC. The receding radius couples transport and geometry, increases the numerical stiffness near the surface as the particle shrinks, and heightens the sensitivity to surface-flux enforcement from external mass transfer. All of these factors are known to expose optimization failure in PINNs (Krishnapriyan et al., 2021). Several strategies have been reported in the literature to alleviate

such issues, including nondimensionalization and variable transformations, time-marching or causality-respecting training, adaptive loss weighting, and feature/basis enrichment to counter spectral bias (Daw et al., 2022; Malekjani et al., 2025b, 2025a; Malekjani et al., 2026; S. Wang, Teng, et al., 2021; Wang, Wang, et al., 2021). For moving boundaries, front-fixing or interface-aware formulations remove domain motion from the computational domain while preserving the mass balance through appropriately transformed operators and BCs (Casabán et al., 2025).

Despite these advances, there remains a need for systematic, reproducible baselines that quantify how combinations of such remedies behave in a single, well-defined moving-boundary transport problem across operating regimes (shrinkage intensity, diffusivity, and particle size). The present study addresses this gap by using a single spherical droplet drying case as a testbed. A comprehensive 2^5 full-factorial screen is designed to identify a progressively improving configuration and to evaluate its robustness against a high-accuracy Crank-Nicolson (CN) reference. The results clarify which mechanisms are most important for reliability for shrinking-domain diffusion and provide a documented baseline for future work on inverse identification and coupled transport.

In this study, the SDD model is restricted to the first drying stage of a spherical binary droplet containing a volatile solvent and a non-volatile solute, that is, to the regime before precipitation, crust formation, or other later-stage structural transitions occur. Shrinkage is described by the d^2 -law, and internal solute redistribution is governed by diffusion in a shrinking domain. This formulation is useful because it isolates the core problem of diffusion in a moving boundary domain under controlled assumptions, and therefore provides a canonical benchmark for assessing the PINN reliability before more complex physics, such as crystallization, shell growth, or coupled heat and mass transfer, are introduced.

The remainder of this paper is structured as follows: Section 2 presents the spatially resolved modeling of single droplet drying, including the governing equations, IC and BCs, and the transformation from a moving to a fixed computational domain. Section 3 introduces the baseline and progressive enhancement path of the PINN framework, detailing the loss terms, sampling strategies, and criteria for optimizer switching. Section 4 reports the results for baseline PINN failure, the progressive enhancement path, and analysis of the effects of each modification relative to the CN reference method. This section also discusses the most appropriate optimizer-switching epoch, validation across multiple drying regimes, the chemical engineering relevance of the model, and a *in silico* demonstration of SDD-PINN for parameter estimation. Finally, Section 5 summarizes the key findings and outlines the directions for future work.

2. Spatially-resolved modeling of single droplet drying

A standard diffusion model is adopted for the first drying stage of a shrinking spherical droplet containing a volatile solvent and a non-volatile solute. During drying, the droplet radius decreases according to the d^2 -law, while the internal solute field evolves by diffusion. This provides a convenient framework for analyzing solute transport in a shrinking droplet under controlled assumptions before additional effects such as crystallization, shell formation, or coupled heat and mass transfer are introduced.

Assuming spherical geometry, the diffusion equation for the species i in the droplet, expressed in terms of mass fraction Y_i , is given by (Brenn, 2004):

$$\frac{\partial Y_i}{\partial t} = \frac{D}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial Y_i}{\partial r} \right), \quad (1)$$

where D is the binary diffusion coefficient of particulates or solids in the liquid. Eq. (1) applies within the radial domain $r \in [0, a(t)]$, where $a(t)$ denotes the time-dependent droplet radius that decreases as drying

progresses. The temporal evolution of $a(t)$ is governed by the rate of shrinkage parameter ($\bar{\alpha} = \frac{da^2}{dt}$), following the d^2 -law, which links the moving boundary to the drying condition.

The governing formulation in Eq. (1) is based on the following assumptions: the model targets the first drying stage of a shrinking solution droplet and assumes spherical symmetry, a single continuous liquid phase, diffusion-dominant internal transport, a non-volatile solute, and prescribed radius evolution through the d^2 -law. The binary diffusion coefficient D is treated as an effective parameter for each benchmark drying condition. Under these assumptions, the formulation resolves internal concentration gradients and surface enrichment on a shrinking domain, but it does not account for internal circulation, concentration-dependent viscosity changes, asymmetric deformation, explicit crystallization, shell/crust growth, or multiphase transport after solid formation. Accordingly, the model should be interpreted as a pre-precipitation / pre-crust formulation, appropriate as long as the droplet remains approximately spherical and the liquid phase remains continuous. Potential solidification is therefore not explicitly modeled in the present work; once local solute enrichment approaches the solubility limit, additional mechanisms such as nucleation, crystal growth, and porous-crust transport would be required.

Coupled heat transfer is not solved explicitly in the present study. Thermal effects therefore enter only indirectly through the selected benchmark shrinkage rates and transport parameters. A more realistic extension would require a coupled heat-and-mass-transfer moving-boundary formulation, in which evaporation, temperature, and radius evolution are determined simultaneously.

The BC at the droplet surface enforces mass conservation between internal transport and interfacial flux and is formulated as follows:

$$-D \frac{\partial Y_i}{\partial r} - \frac{da}{dt} Y_i = \frac{\dot{m}_i}{A \rho_l}, \quad (2)$$

where A is the droplet surface area, ρ_l is the liquid density and $\dot{m}_i$ the mass transfer rate of species i across the droplet surface into the surrounding gas phase. The first term represents diffusion within the liquid droplet, the second term accounts for the convective contribution due to the receding interface, and the right-hand side denotes the mass flux leaving the droplet surface. For a non-volatile solute, the latter term is zero. At the center of the droplet, the regularity condition must be satisfied, i.e., the radial gradient of the species mass fraction profile is zero for all times t :

$$\frac{\partial Y_i}{\partial r} = 0. \quad (3)$$

The IC is specified as follows:

$$Y_i(t=0, r) = Y_{i0}(r). \quad (4)$$

To transform the moving spherical domain to a fixed computational domain, the variable transformation $\xi = r/a(t)$ was applied following Brenn (2004). Accordingly, Eq. (1) was converted to a nonlinear advection-diffusion PDE using the front-fixing method (details provided in Appendix A):

$$a^2 \frac{\partial Y_i}{\partial t} = \frac{da^2}{dt} \frac{\xi}{2} \frac{\partial Y_i}{\partial \xi} + \frac{D}{\xi^2} \frac{\partial}{\partial \xi} \left(\xi^2 \frac{\partial Y_i}{\partial \xi} \right). \quad (5)$$

This transformation converts the moving spatial domain $r \in [0, a(t)]$ to a fixed domain $\xi \in [0, 1]$. The first term at the right-hand side of Eq. (5) is an advection-drift term that appears as a consequence of spatial domain transformation and is responsible for accounting for shrinkage of the domain.

The surface BC, center regularity condition, and IC transform accordingly. Eq. (2) becomes:

$$-D \frac{\partial Y_i}{\partial \xi} - \frac{\bar{\alpha}}{2} Y_i = \frac{\dot{m}_i}{4\pi a \rho_l}, \quad (6)$$

The regularity condition at $\xi = 0$ is:

$$\frac{\partial Y_i}{\partial \xi} = 0, \quad (7)$$

and the IC becomes:

$$Y_i(t=0, \xi) = Y_{i0}(\xi). \quad (8)$$

The problem can then be nondimensionalized with respect to droplet lifetime t_l defined as:

$$t_l = \frac{a_0^2}{\bar{\alpha}} \left[\left(\frac{\bar{Y}_{20}}{(1 - \bar{Y}_{20}) \rho_{s2} / \rho_l + \bar{Y}_{20}} \right)^{2/3} - 1 \right], \quad (9)$$

where a_0 is the initial droplet radius, $\bar{Y}_{20}$ is the mean initial mass fraction of the solute in the droplet, and ρ_{s2} is solid density. In Eq. (9) it is assumed that, at the end of the drying process, the particle density is equal to the compact solid density ρ_{s2} . The nondimensional form of Eq. (5) is therefore:

$$\bar{a}^2 \frac{\partial Y_i}{\partial \tau} = \alpha \frac{\xi}{2} \frac{\partial Y_i}{\partial \xi} + \frac{G}{\xi^2} \frac{\partial}{\partial \xi} \left(\xi^2 \frac{\partial Y_i}{\partial \xi} \right), \quad (10)$$

where

$$\bar{a} = \frac{a}{a_0}, \quad \tau = \frac{t}{t_l}, \quad \alpha = \frac{\bar{\alpha} t_l}{a_0^2}, \quad G = \frac{t_l D}{a_0^2}.$$

The BC (Eq. (6)) is rewritten as:

$$-\frac{\partial Y_i}{\partial \xi} - \frac{\bar{\alpha}}{2D} Y_i = \frac{\dot{m}_i}{4\pi a D \rho_l}, \quad (11)$$

and the IC becomes:

$$Y_i(\tau=0, \xi) = Y_{i0}(\xi). \quad (12)$$

If the droplet evaporates according to the d^2 -law, its nondimensional form reads as:

$$\frac{a^2(t)}{a_0^2} = 1 + \alpha \tau. \quad (13)$$

Defining

$$s = -\frac{\alpha}{4G}, \quad (14)$$

Eq. (10) becomes:

$$\frac{\partial Y_i}{\partial \tau} = \frac{G}{1 + \alpha \tau} \left[\frac{\partial^2 Y_i}{\partial \xi^2} + \left(\frac{2}{\xi} - 2s\xi \right) \frac{\partial Y_i}{\partial \xi} \right]. \quad (15)$$

Since no evaporation takes place for the solute (species 2), $\dot{m}_2 = 0$ and Eq. (11) becomes:

$$-\frac{\partial Y_2}{\partial \xi} = \frac{\alpha}{2G} Y_2. \quad (16)$$

As in Brenn (2004), we use a polynomial function for the IC as:

$$Y_{20}(\xi) = c_0 + c_n \xi^n. \quad (17)$$

This is the simplest form that satisfies both the center and surface BCs. The coefficients c_0 and c_n are determined by enforcing symmetry at the droplet center and the surface BC, together with the requirement that the integral of the distribution $Y_{20}(\xi)$ over the droplet volume must yield $\bar{Y}_{20}$ times the initial droplet volume as:

$$c_0 = \bar{Y}_{20} \frac{n + \frac{\alpha}{2D}}{n + \frac{n+3}{2} \frac{\alpha}{2D}}, \quad (18)$$

$$c_n = -\bar{Y}_{20} \frac{\frac{\tilde{a}}{2D}}{n + \frac{n}{n+3} \frac{\tilde{a}}{2D}}, \quad (19)$$

where $\bar{Y}_{20}$ is the known overall mass fraction of the solute in the droplet at $\tau = 0$, and n is taken to be 5.

The equations above were solved using an implicit CN method with 401 grid points in the spatial domain ($\Delta\xi \approx 0.0025$) and 900 uniform time steps. These settings provided numerically stable and convergent reference solutions for comparison with the PINNs.

The material system considered here is an aqueous sodium chloride solution, consistent with the benchmark case reported by Brenn (2004). In the reference case, the droplet shrinks at a constant rate $\tilde{a} = -0.46 \times 10^{-9}$ (m²/s). The binary diffusion coefficient is taken as $D = 1.61 \times 10^{-9}$ (m²/s). The initial droplet radius is $a_0 = 0.734$ mm and the initial average solute mass fraction is $\bar{Y}_{20} = 0.06$. The densities of the solvent ρ_1 and solute ρ_{s2} are 1000 and 2165 kg/m³, respectively. This drying regime is hereafter referred to as the reference regime. The present formulation is therefore interpreted within the corresponding pre-saturation regime; once the surface concentration approaches this limit, precipitation/crust-related physics would need to be introduced which is beyond the scope of the current study.

To assess robustness, we re-trained the SDD-PINN across six drying regimes (Table 1) spanning droplet size, rate of droplet shrinkage, and binary diffusion coefficient. To quantify the relative importance of droplet shrinkage and solute diffusion across regimes, the corresponding Péclet number is reported as $Pe = -\tilde{a}/2D$, where larger Pe indicates stronger shrinkage effects relative to diffusion. It should be noted that changes in D in the “slow diff” and “fast diff” cases should be interpreted as deliberate stress-test settings rather than as temperature-driven variations in material properties.

3. Physics-informed neural networks

Physics-informed neural networks were used to solve Eq. (15) incorporating the appropriate IC and BCs at the surface and center to predict the concentration gradient profiles of solute inside the droplet. PINNs use fully connected feed-forward neural networks to solve the equation. The network consists of multiple hidden layers and maps τ and ξ to Y_2 . The weights and biases are gradually updated iteratively during the training process until the network parameters converge. Four loss terms are minimized during PINN training, penalizing deviations in the dynamics, the regularity, the surface BC, and the IC, respectively:

$$Loss_{PDE} = \frac{1}{N_{PDE}} \sum_{j=1}^{N_{PDE}} \left(\frac{\partial Y_i}{\partial \tau} - \frac{G}{1 + \alpha \tau} \left[\frac{\partial^2 Y_i}{\partial \xi^2} + \left(\frac{2}{\xi} - 2s\xi \right) \frac{\partial Y_i}{\partial \xi} \right] \right)^2, \quad (20)$$

Table 1
Different drying regimes for validation of SDD-PINN performance.

Drying regime	a_0 (mm)	$\tilde{a}$ (m ² /s)	D (m ² /s)	Peclet number	Definition
High	0.734	-5.63×10^{-9}	1.61×10^{-9}	1.75	High rate of droplet shrinkage
Very high	0.734	-7×10^{-9}	1.61×10^{-9}	2.17	Very high rate of droplet shrinkage
Mid	0.734	-2×10^{-9}	1.61×10^{-9}	0.62	Medium rate of droplet shrinkage
Small a_0	0.5	-2×10^{-9}	1.61×10^{-9}	0.62	Small droplet radius
Slow diff	0.734	-2×10^{-9}	0.80×10^{-9}	1.25	Small binary diffusion coefficient
Fast diff	0.734	-2×10^{-9}	3.22×10^{-9}	0.31	High binary diffusion coefficient

$$Loss_{center} = \frac{1}{N_{center}} \sum_{k=1}^{N_{center}} \left(\frac{\partial Y_i}{\partial \xi} \Big|_{\xi=0} \right)^2, \quad (21)$$

$$Loss_{surface} = \frac{1}{N_{surface}} \sum_{\ell=1}^{N_{surface}} \left(\left(-\frac{\partial Y_i}{\partial \xi} - \frac{\alpha}{2G} Y_i \right) \Big|_{\xi=1} \right)^2, \quad (22)$$

$$Loss_{IC} = \frac{1}{N_{IC}} \sum_{m=1}^{N_{IC}} (Y_i|_{\tau=0} - Y_{20})^2, \quad (23)$$

where N_{PDE} , N_{center} , $N_{surface}$, and N_{IC} denote the number of collocation points used to enforce the PDE residual, center regularity condition, surface BC, and initial concentration profile, respectively. The indices j , k , ℓ , and m enumerate these collocation points in their corresponding sets.

Combining all losses (Eqs. (20)-(23)), the total loss function is defined as:

$$Loss_{total} = \omega_{PDE} Loss_{PDE} + \omega_{center} Loss_{center} + \omega_{surface} Loss_{surface} + \omega_{IC} Loss_{IC}, \quad (24)$$

where ω is the weight of each loss component. A schematic representation of the baseline PINN used in this study is presented in Fig. 1. Details of the baseline PINN configuration and training settings are summarized in Table 2.

In Section 2, the moving-boundary PDE (Eq. (1)) has been transformed to a fixed domain form (Eq. (5)). The corresponding change in the sampling domain relevant to the PINN implementation is illustrated in Fig. 2.

3.1. Progressive enhancement path

In order to enhance the PINN's performance, a compact, physics-motivated PINN recipe is proposed:

- (i) Hard enforcement of the IC, which removes the IC loss term and its gradient competition. This turns the IC from a quantity that must be enforced through optimization into a structural constraint, thereby simplifying the optimization process.
- (ii) A squared transformed radius (ξ^2) is used as an alternative spatial input to introduce an inductive bias consistent with spherical center symmetry. In Eq. (15), the transformed operator contains the $2/\xi$ term, so the representation near $\xi=0$ is especially sensitive to how the radial coordinate is encoded. The ξ^2 input does not replace the center BC; it acts as an inductive bias that favors center-regular representations.
- (iii) A log-time reparameterization is performed as:

$$\theta = -\ln(1 + \alpha \tau). \quad (25)$$

Substituting this transformation into Eq. (15) yields (Appendix B):

$$\frac{\partial Y_i}{\partial \theta} = -\frac{G}{\alpha} \left[\frac{\partial^2 Y_i}{\partial \xi^2} + \left(\frac{2}{\xi} - 2s\xi \right) \frac{\partial Y_i}{\partial \xi} \right]. \quad (26)$$

This remapping converts the algebraic decay $\frac{G}{1+\alpha\tau}$ in Eq. (15) into a constant factor $-\frac{G}{\alpha}$. Time dependence does not disappear in general; it is just absorbed by this mapping. This reformulation balances coefficients and makes PINN learning easier.

- (iv) Smart collocation sampling, which first draws pseudo-random points distributed across the space-time domain (and, when shrinkage is strong, it places additional points near the particle surface) to ensure broad coverage.

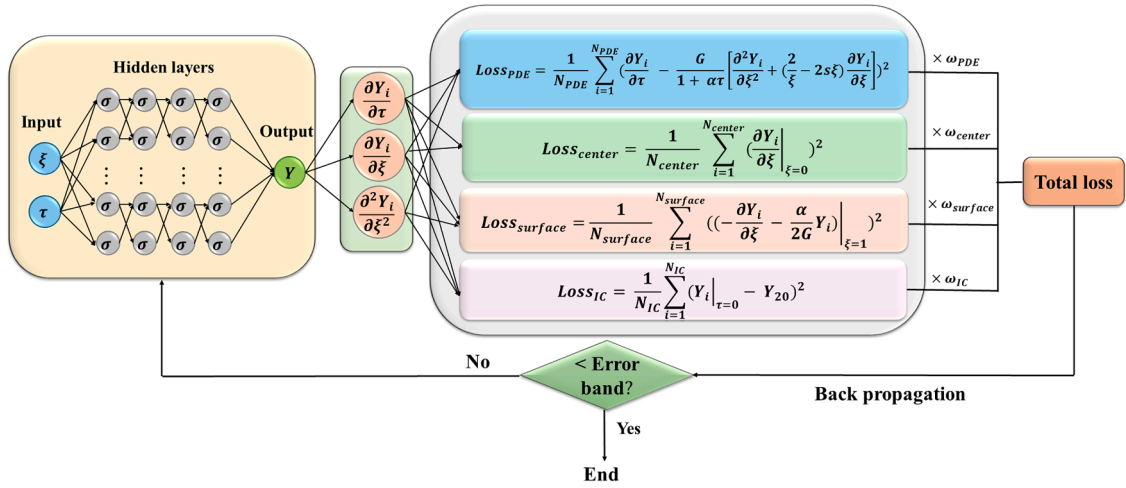


Fig. 1. Schematic of baseline PINN used to solve the SDD problem.

Table 2
PINN configuration and training settings.

Item	Setting
Network	Multilayer perceptron (MLP), 6 layers (5 hidden, 1 output), 128 neurons
Activation function	tanh (all hidden), last layer linear
Initialization	Xavier/Glorot uniform for weights, zeros for biases
Collocation sampling (Uniform sampling)	$N_{PDE} = 16,000$, $N_{IC} = 1,200$, $N_{center} = 1,200$, $N_{surface} = 3,600$
Optimizer	Adam at an initial learning rate of 0.001 with a learning rate scheduler
Epochs / resampling	2000 epochs; resample collocation every 600 epochs
Loss metrics	Mean squared error (MSE) of the residuals
Software & libraries	PyTorch 2.1.0 (Python 3.9.18) on CUDA 11.8 / cuDNN 8.7
Device	GPU: NVIDIA RTX A4500 (20 GB)

It then adds more points exactly in regions where the PDE residual is largest and refreshes this set during training. Thus, sampling progressively focuses on the hardest regions while still keeping overall coverage.

(v) A combined Adam + L -BFGS optimization schedule is used. Adam quickly finds a good basin, while L -BFGS exploits curvature information to improve the rate of residual reduction. This hybrid optimization is now widely used in PINN literature to reduce final errors

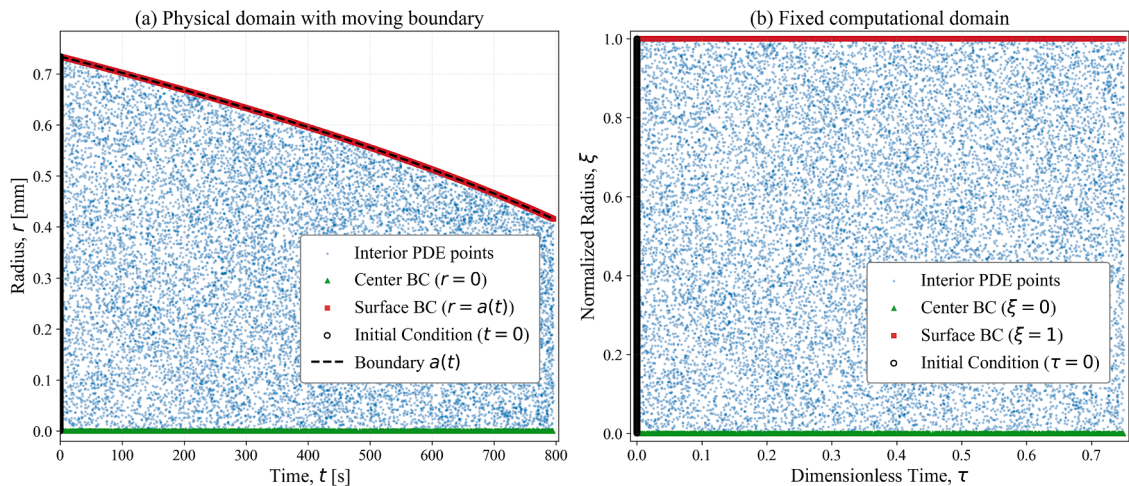
significantly (Lan et al., 2024; Shen et al., 2025; Turan and Dutta, 2025).

A 2^5 full-factorial design of experiments (soft vs. hard IC, ξ vs. ξ^2 , τ vs. θ , uniform vs. smart sampling, Adam vs. Adam + L -BFGS) is first performed on the reference drying regime to identify a progressive enhancement path. The selected configuration, with the lowest mean relative L_2 error (mean relL2) against a CN reference, was verified across multiple drying conditions spanning shrinkage intensity, binary diffusion coefficient, and particle size with three repeated runs using different random seeds. At each evaluation time τ_k , the relative L_2 error between the PINN predictions and CN reference values was calculated as:

$$\varepsilon(\tau_k) = \frac{\|Y(\xi, \tau_k) - Y_{ref}(\xi, \tau_k)\|_2}{\|Y_{ref}(\xi, \tau_k)\|_2} \quad (27)$$

The mean relL2 is then calculated as the average over all K evaluation times, where K is the number of discrete time points used for CN-PINN comparison in each drying condition:

$$Mean \text{ relL2} = \frac{1}{K} \sum_{k=1}^K \varepsilon(\tau_k). \quad (28)$$

Fig. 2. Collocation sampling (a) before, and (b) after the $\xi = r/a(t)$ transformation.

3.2. Optimizer selection and switching criteria

As mentioned in Section 3.1, switching from Adam to the L-BFGS optimizer can significantly improve performance. To keep training economical, after establishing the predictive ability of SDD-PINN (the best-performing PINN in Section 3.1), we designed a two-stage criterion to avoid trivial Adam epochs and the hybrid strategy was then used for all results reported here.

The selection criteria are presented in Fig. 3. First, in a single-seed screen, we train using Adam for 6000 epochs and “lock” snapshots every 500 epochs. At each lock we run a short L-BFGS from the snapshot and score against the CN reference. A lock is a candidate only if it (i) passes an accuracy guardrail (mean relL2 ≤ 0.025), (ii) is economical, meaning that the next lock ($L + 500$) would not improve mean relL2 by $>3\%$, and (iii) passes a non-regression check requiring no $>10\%$ degradation versus the previous lock. We take the earliest lock that satisfies these checks (or the best available fallback). A 1000-epoch warm-up for all runs was considered to avoid early switching.

After selecting the lock, a multi-seed confirmation was performed to ensure reproducibility. We reran only the candidate locks across multiple seeds with a longer L-BFGS and require reproducibility: for a candidate epoch, for a candidate epoch, the fraction of seeds whose mean relL2 falls within $\pm 10\%$ of that seed’s own best value must exceed a threshold (here, ≥ 0.70), and its median mean relL2 must be non-inferior to the best candidate within a small margin δ (e.g., 0.0015). We then choose the earliest epoch that meets both reproducibility and non-inferiority; if none does, we fall back to the candidate with the lowest median mean relL2.

4. Results and discussion

4.1. The failure of the baseline PINN

The baseline PINN (Fig. 1) does not provide a reliable solution for the shrinking droplet drying problem. As shown in Table 3, the mean relL2 error of the baseline PINN is already substantial in the reference regime and remains large across all tested regimes. The mean relL2 error ranges from 0.463 ± 0.059 in the reference regime to 0.961 ± 0.001 in the very-high regime, while similarly large errors are also observed for other cases. This shows that the baseline PINN failure is not restricted to a single operating point. Rather, it persists across the full tested parameter set and is therefore consistent with a structural limitation of the baseline PINN formulation for this class of problems. The small standard deviations in the stiffer regimes further indicate that this poor performance is systematic across seeds rather than an artifact of unfavorable initialization.

Table 3

Results of baseline PINN at different drying regimes.

Regime	Mean relL2	RMSE
Reference (low)	0.463 ± 0.059	$1.244 \times 10^{-1} \pm 8.251 \times 10^{-3}$
Mid	0.769 ± 0.007	$9.284 \times 10^{-2} \pm 8.148 \times 10^{-4}$
High	0.949 ± 0.001	$9.856 \times 10^{-2} \pm 1.112 \times 10^{-4}$
Very high	0.961 ± 0.001	$6.435 \times 10^{-2} \pm 4.373 \times 10^{-5}$
Slow diff	0.914 ± 0.002	$9.913 \times 10^{-2} \pm 2.839 \times 10^{-4}$
Fast diff	0.559 ± 0.016	$7.875 \times 10^{-2} \pm 1.959 \times 10^{-3}$
Small a_0	0.698 ± 0.002	$5.175 \times 10^{-2} \pm 1.680 \times 10^{-4}$

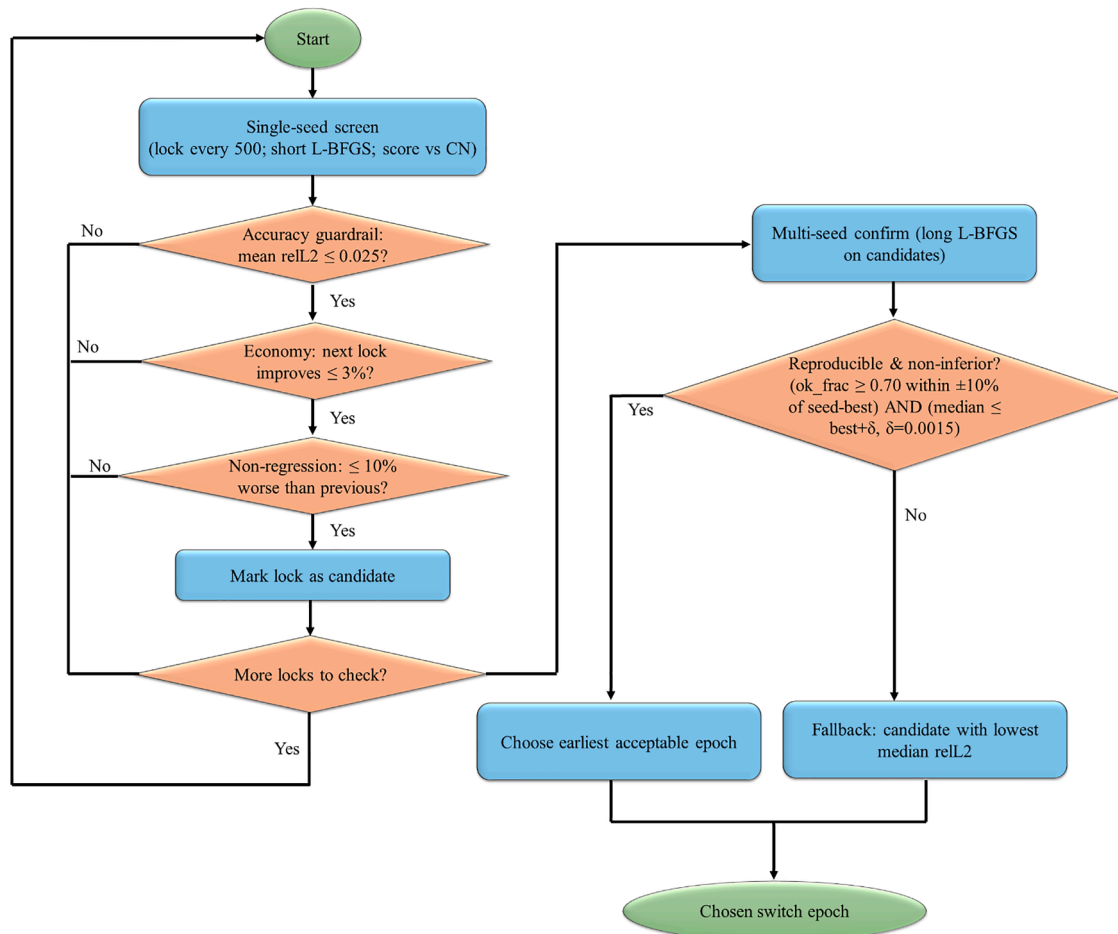


Fig. 3. Optimizer switch from Adam to L-BFGS selection criteria.

Fig. 4 complements this by showing how the baseline error (grey curve, Step 0) evolves over drying time. The error propagation curve shows that the baseline PINN is relatively more accurate near the initial time but loses predictive accuracy as drying proceeds. This behavior shows that the PINN fails to propagate the correct transport dynamics over time. In other words, the model does not diverge numerically, but it fails in terms of quantitative accuracy and exhibits a loss of temporal predictive fidelity.

To investigate the underlying mechanisms, we employ training diagnostics inspired by multitask deep learning and PINN gradient analyses (Chen et al., 2018; Wang et al., 2021). The detailed histories of these diagnostics are provided in Appendix C. These diagnostics include: (1) residual imbalance, which tracks the uneven reduction of the PDE, IC, and BC loss components (Eqs. (20)–(23)) over epochs; (2) gradient norms, computed with respect to the trainable network parameters, which quantify how strongly each loss component influences the optimization updates; and (3) operator-term diagnostics, which resolve Eq. (15) (or Eq. (26)) into root mean square (RMS) magnitudes of the main components: the temporal derivative, the diffusion term, the drift-like first-derivative term, and the full coefficient-weighted spatial operator on the right-hand side (i.e., the complete spatial contribution appearing on the right-hand side).

The diagnostics in Appendix C clarify the origin of the baseline failure. As shown in Fig. C1, the PDE gradient dominates the IC and BC gradients, so the optimization updates remain highly unbalanced even when the loss values themselves appear moderate. The residual histories in Fig. C2 show the same pattern: the IC and boundary-related terms are reduced comparatively easily, whereas the PDE residual remains much harder to optimize over the evolving time domain. Together, these results indicate that the baseline PINN learns the easier constraints first, but fails to resolve the interior evolution with comparable accuracy.

Fig. C3 provides additional mechanistic insight. In the baseline τ -formulation, the spatial operator is multiplied by the time-dependent factor $G/(1+\alpha\tau)$, which increases as drying proceeds and amplifies the effective stiffness of the transformed PDE. The monitored RMS magnitudes therefore show that the scaled operator remains persistently large during training, indicating that the temporal-spatial balance becomes progressively harder to maintain. The baseline network thus struggles with the coupled effects of diffusion, geometry change, and temporal evolution in the moving domain.

Taken together, these results show that the failure of the vanilla baseline PINN arises from a combination of regime-wide breakdown (Table 3), rapid late-time error growth (Fig. 4), strong PDE-dominated

gradient imbalance (Fig. C1), PDE residuals that remain difficult to reduce relative to IC/BC terms (Fig. C2), and stiffness amplification in the transformed operator (Fig. C3). The baseline therefore serves as a useful negative control, motivating the progressive enhancements introduced in Section 4.2.

4.2. Progressive enhancement path for selected drying conditions

The 2^5 full-factorial plan (Fig. 5) was first evaluated in the reference drying regime in order to identify a progressive enhancement path under controlled conditions, as introduced in Section 3.1. Fig. 6 shows the improvements through the progressive enhancement path. At Step 0, as described in Section 4.1, the predicted profiles remain too close to the initial state while underpredicting the concentration build-up at later drying times (Fig. 7a). At Step 1, only the IC treatment is changed, i.e., the baseline model is replaced by its hard-IC counterpart. This removes the need for the optimizer to learn the IC through a competing soft penalty and therefore eliminates one source of early training competition. As a result, the mean rell2 decreases from 0.463 ± 0.059 to about 0.298 ± 0.015 . Fig. C1 shows, however, that hard IC does not resolve the dominant pathology by itself: the IC loss/gradient channel disappears, but the PDE gradient remains strongly dominant and the PDE residual is still the slowest component to decrease. Step 1 therefore improves accuracy mainly by removing unnecessary early competition with the IC, while leaving the main interior-evolution difficulty and the late-time bias largely unresolved.

At Step 2, the spatial input is changed from ξ to ξ^2 , and the optimizer is upgraded from Adam-only to Adam + ι -BFGS. The mean rell2 decreases further to about 0.140 ± 0.026 , corresponding to an improvement of about 53 % relative to Step 1 (Fig. 6), and the concentration profiles become visibly more accurate (Fig. 7c). In the gradient histories (Fig. C1), the imbalance between interior- and boundary-driven optimization signals is reduced compared to Step 1, while the loss/residual histories (Fig. C2) show further tightening of the PDE and surface terms. The optimizer switch is also important: once Adam has reached a reasonable parameter region, the subsequent ι -BFGS stage provides additional late-stage reduction of the remaining residuals, as seen after the optimizer transition in Figs. C1. Thus, the main gain at Step 2 is most consistently explained as a synergistic effect: ξ^2 improves the conditioning of the spatial representation, while ι -BFGS becomes more effective on the resulting optimization landscape.

At Step 3, the original τ -formulation is replaced by the logarithmic time reparameterization θ , while retaining hard IC, the ξ^2 input, and the

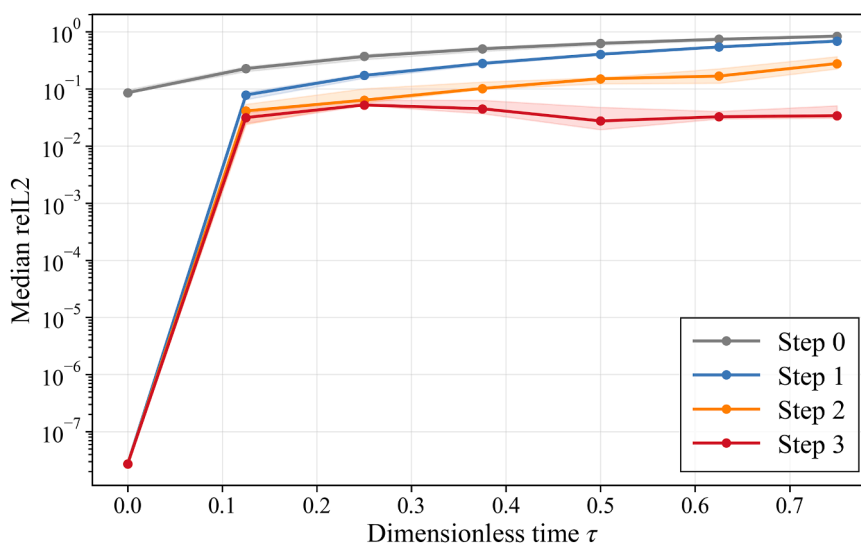


Fig. 4. Error propagation of the reference drying regime in the progressive enhancement path.

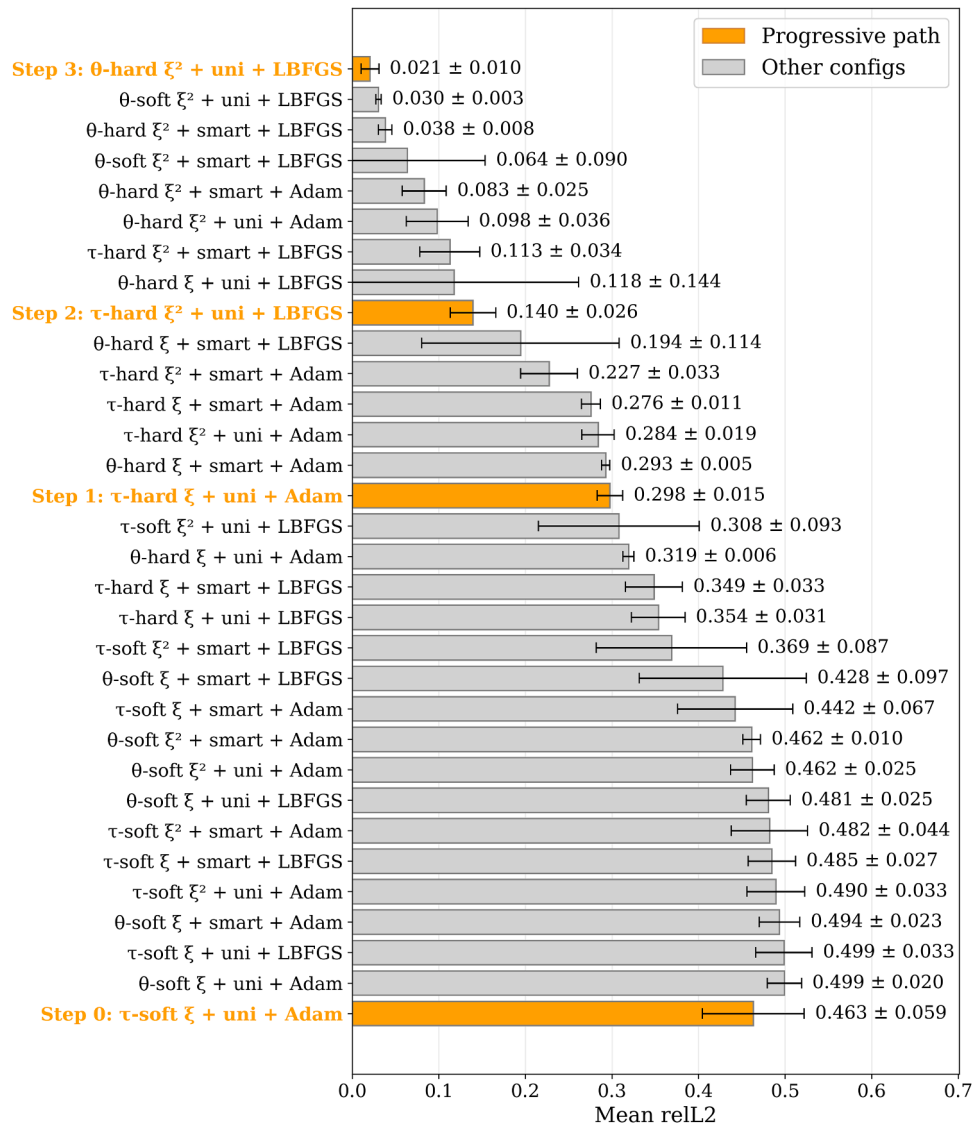


Fig. 5. Full-factorial grid error showing the selected progressive enhancement path (hard and soft denote hard and soft IC, ξ and ξ^2 indicate the spatial input feature, uni and smart correspond to uniform and smart sampling).

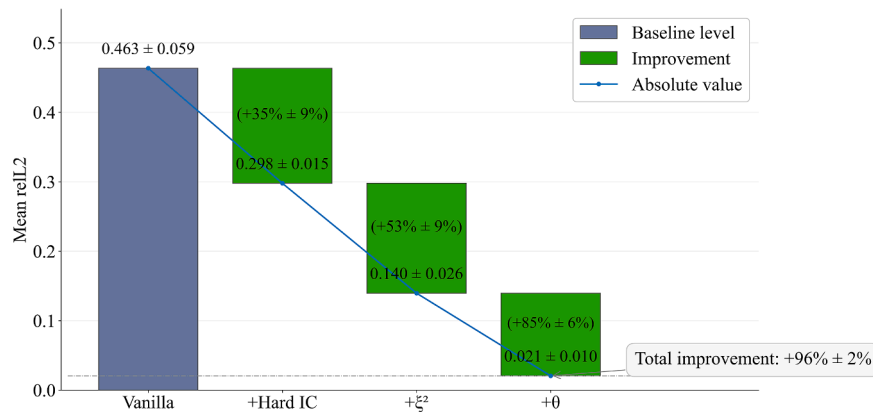


Fig. 6. Stepwise improvement of error through progressive enhancement path.

Adam + L -BFGS optimizer. This yields the largest improvement, with mean relL2 dropping to 0.021 ± 0.010 , corresponding to an improvement of about 85 % relative to Step 2 and about 96 % relative to the

baseline (Fig. 6). The concentration profiles at this step match the CN reference closely over the full drying interval (Fig. 7d). The operator-term diagnostics in Appendix C clarify why this step is especially

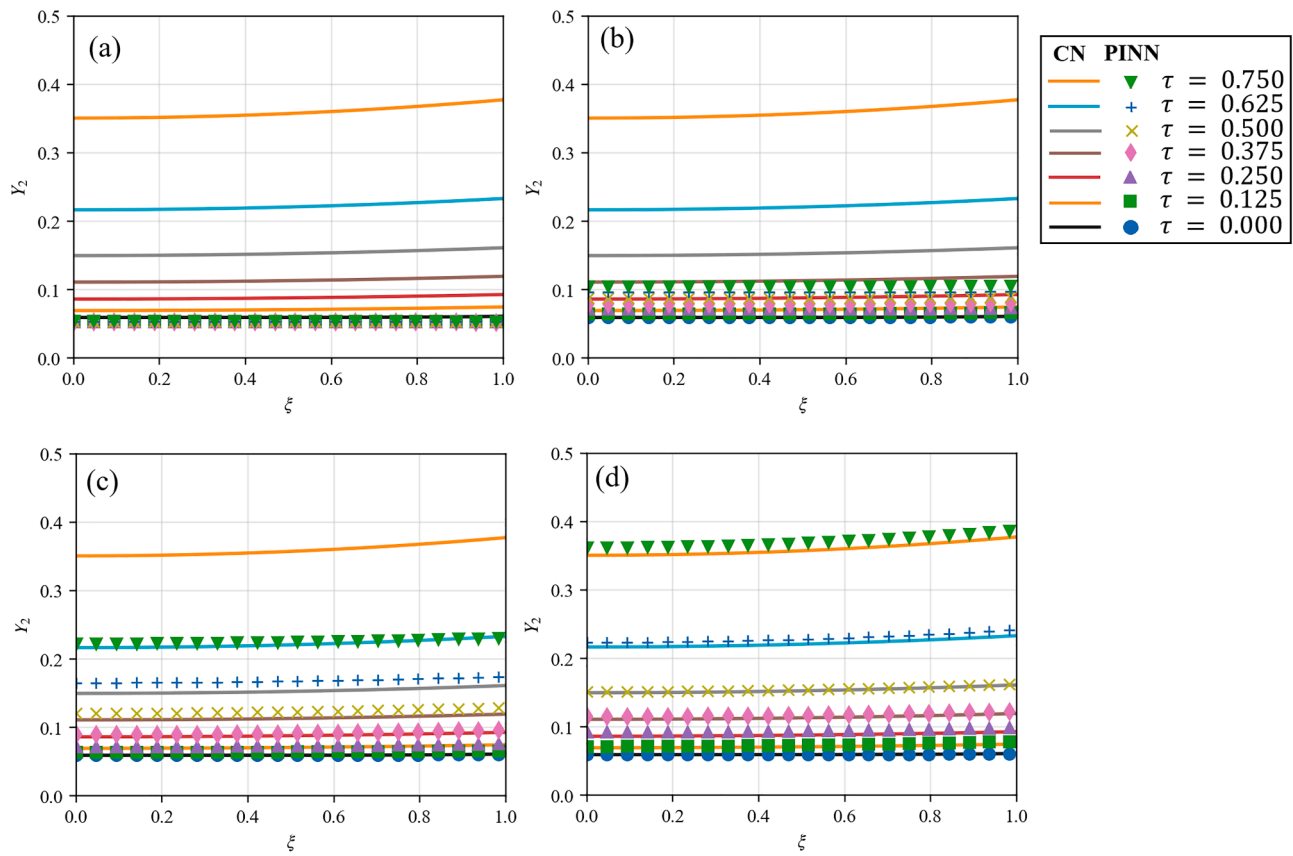


Fig. 7. Concentration profiles predicted by the PINN for the reference drying regime (a) step 0, (b) step 1, (c) step 2, (d) step 3.

effective. In the original τ -formulation, the spatial operator is multiplied by the time-dependent factor $G/(1+\alpha\tau)$, which increases as drying proceeds and therefore amplifies the effective stiffness of the transformed PDE at later times. The θ -transformation removes this growing prefactor by absorbing it into the new time coordinate, recasting the problem into a form with much more uniform operator scaling over the drying interval. In the diagnostics in [Appendix C](#), this effect is reflected more clearly in the overall training behavior than in any single operator component alone: compared with Step 2, Step 3 shows stronger suppression of late-time error growth ([Fig. 4](#)) and a more pronounced final-stage reduction of the PDE and surface residuals after the optimizer transition ([Fig. C2d](#)).

Overall, the progressive path is therefore mechanistically interpretable. Hard IC removes unnecessary early competition from the IC loss term, ξ^2 improves representational alignment with the spherical radial structure, Adam + ι -BFGS improves final-stage optimization, and θ mitigates the late-time stiffness amplification intrinsic to the τ -formulation. Together, these modifications transform the inaccurate vanilla baseline into the robust Step 3 configuration, referred to hereafter as SDD-PINN. Within the full-factorial study, smart sampling did not provide additional benefit for the final recipe and was therefore not retained in subsequent experiments. This indicates that, for the present problem, the dominant limitation is not insufficient collocation density, but the combined coordinate, temporal, and optimization imbalance described

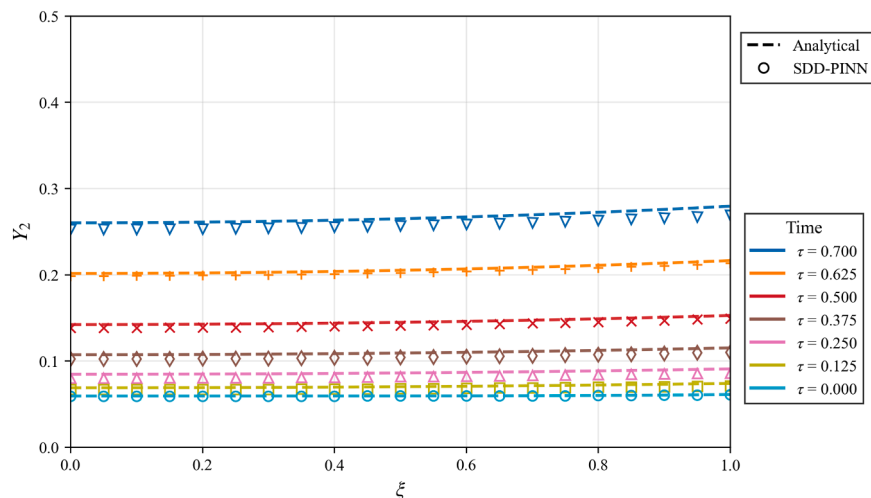


Fig. 8. Comparison of concentration profiles predicted by SDD-PINN with the analytical solution of [Brenn \(2004\)](#) for the reference drying regime.

above.

An additional model consistency check was carried out against the corresponding analytical solution adapted from Brenn (2004). Fig. 8 compares the final SDD-PINN in the reference drying regime with the analytical benchmark. The SDD-PINN predictions follow the analytical solution closely over the full radial domain and drying interval. This shows that the final SDD-PINN reproduces not only the numerical CN reference used throughout the paper, but also an analytical solution for the simplified first-stage case. It should be noted that this comparison does not replace experimental validation, which remains beyond the scope of the present benchmark model.

Fig. 9 summarizes pairwise interactions on the error metrics, highlighting when improvements are super-additive (positive synergy) versus merely additive or antagonistic. The strongest positive synergistic effect appears between ξ^2 and combined Adam + L-BFGS optimizer (+0.160), and between time reparameterization to θ and combined Adam + L-BFGS optimizer (+0.133). This pattern shows that L-BFGS is more beneficial when paired with ξ^2 and θ . Conversely, applying L-BFGS to poorly conditioned setups offers limited benefit. Hard-IC shows only weak interactions (slightly positive with time reparameterization to, slightly negative with the combined Adam + L-BFGS optimizer), consistent with its role as an early-time stabilizer that mainly removes loss imbalance rather than reshaping the overall optimization basin. When the main conditioning remedies are in place, especially θ together with ξ^2 , smart sampling adds little and may even slightly degrade performance. Together, the heatmap supports the practical recipe: first improve the conditioning (ξ^2 , θ), then exploit L-BFGS.

4.3. Optimal epoch for optimizer switch

Based on the procedure in Section 3.2, the optimal epoch for switching from Adam to the L-BFGS optimizer, using the guardrail and multi-seed reproducibility criteria of Fig. 3, was determined. Fig. 10 shows the SDD-PINN performance on the reference drying regime for Adam versus the combined Adam + L-BFGS optimizer. Across all evaluated epochs, the combined optimizer yielded substantially lower mean reLL2 error than Adam alone. The earliest point satisfying both selection

criteria is epoch 2000 (Fig. 10), which we adopt as an economical yet reliable switch point in this study. The corresponding loss trajectory (Fig. 11) shows the Adam period plateauing near the switch, followed by a pronounced reduction in total training loss during the subsequent L-BFGS period.

4.4. Validation under different drying conditions and extreme cases

Across all regimes, the regime-averaged mean reLL2 (vs. CN reference) ranged from 2.69×10^{-3} to 3.30×10^{-2} (Fig. 12). In addition, the mean reLL2 curves (evaluated at each dimensionless time and averaged across seeds) in Fig. 12(b) show that the error remains low throughout the drying process for all regimes, although its magnitude and temporal evolution depend on the operating condition. The largest deviations are observed for the fast-diff and high regimes, whereas the very high and small a_0 regimes maintain the lowest error levels over the full-time range. Importantly, no regime shows uncontrolled late-time error amplification. Fig. 13 presents the concentration profiles predicted by SDD-PINN for different drying regimes, together with the CN reference. As shown, a close agreement is achieved between SDD-PINN predictions and CN results, indicating the method is robust across the tested drying regimes.

4.5. Chemical-engineering relevance and practical implications

Beyond numerical accuracy, the value of SDD-PINN lies in its ability to resolve the evolving intra-droplet concentration field in a form directly useful for drying analysis. Within the present first-stage, single-phase formulation, the most relevant outputs are the radial concentration gradients and the near-surface concentration history, which are not available from droplet-averaged models alone. These quantities describe how internal diffusion competes with interface recession and thus provide the transport information that precedes later concentration-driven transitions. Accordingly, the present model should not be interpreted as a direct predictor after crust formation, locking, or final particle morphology.

At the process scale, accurate single-droplet solutions can serve as

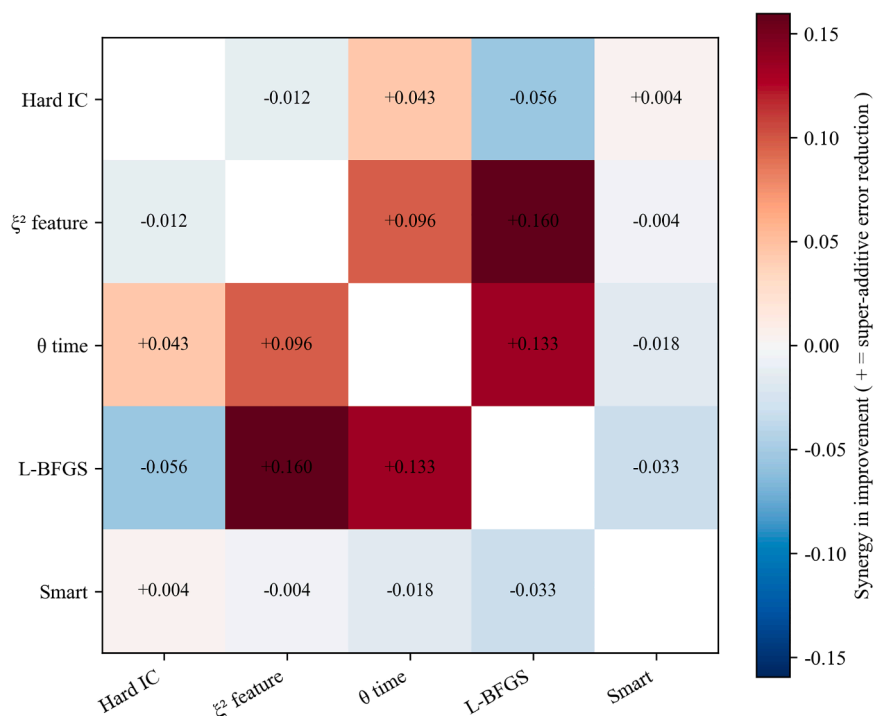


Fig. 9. Pairwise interactions on error metrics.

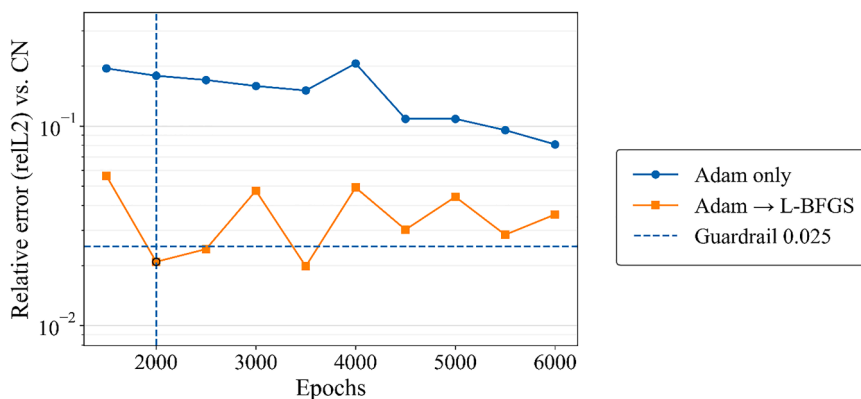


Fig. 10. Relative error of Adam only and combined Adam + L-BFGS optimizer.

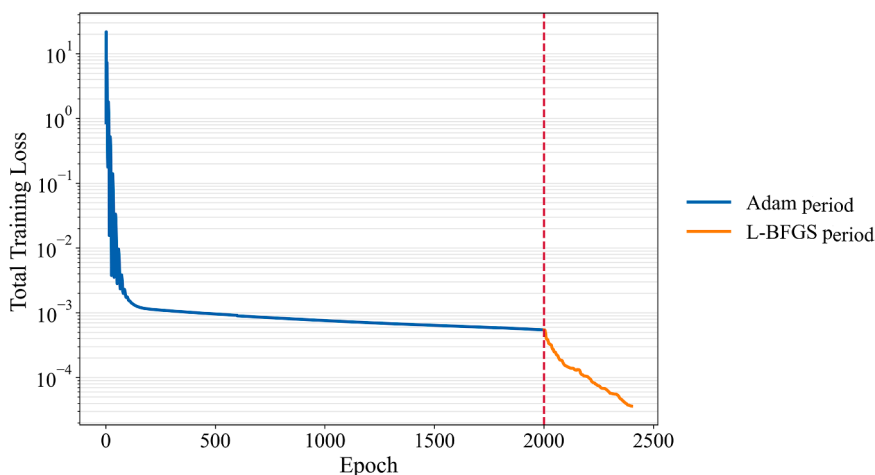


Fig. 11. Loss reduction for the trained SDD-PINN in the reference drying regime using the combined Adam + L-BFGS optimizer.

reference data for reduced closures in spray-dryer CFD, where solving the full intra-droplet PDE for every tracked particle is generally impractical. The same physics-constrained formulation is also attractive for inverse use, because sparse measurements can be linked back to effective transport parameters and the unobserved internal state, as illustrated in Section 4.6.

4.6. In silico demonstration of SDD-PINN for parameter estimation

To illustrate that the proposed framework is not limited to forward prediction, a synthetic inverse test was performed for simultaneous recovery of the binary diffusion coefficient, D , and the dimensional rate of shrinkage parameter, $\tilde{\alpha}$, from sparse noisy observations. This example is intended as a proof of concept rather than a full inverse study. Its purpose is to examine whether the stabilized SDD-PINN formulation can infer unknown drying parameters while also reconstructing the underlying concentration field in a physically consistent manner.

The inverse demonstration was constructed in a model-consistent way using the reference drying regime. First, synthetic reference data were generated with the high-accuracy CN solver using the true parameter values from the reference drying regime. From this reference solution, three sparse observable sets were extracted: (i) the droplet radius history (t), (ii) the bulk-average concentration $\bar{Y}(t)$, and (iii) a surface concentration $Y_s(t) = Y(\xi = 1, t)$. Relative Gaussian noise was then added to emulate measurement uncertainty, with noise levels of 0.5 % for the radius data and 1 % for the concentration-related observables.

For the inverse model, the same stabilized SDD-PINN structure identified in the forward study was retained. The unknown parameters D

and $\tilde{\alpha}$ were embedded directly into the computational graph and optimized jointly with the network weights. Positivity of D and negativity of $\tilde{\alpha}$ were enforced by exponential reparameterization. The total loss combined the physics residual, the transformed BCs, the mismatch to the bulk-average and surface concentration, the mismatch to the radius history, and an additional domain-admissibility penalty to ensure physically meaningful shrinkage over the observation interval.

Starting from the perturbed initial guesses $D = 9 \times 10^{-10} \text{ (m}^2/\text{s)}$ and $\tilde{\alpha} = -3 \times 10^{-10} \text{ (m}^2/\text{s)}$, the corresponding final relative errors were 3.56 % for D and 1.17 % for $\tilde{\alpha}$. These results show that the framework can recover both the transport parameter and the shrinkage parameter with good accuracy from sparse noisy measurements. The observable fits are shown in Fig. 14. The parameter-convergence histories are presented in Appendix D, Figs. D1 and D2.

Beyond matching the measured observables, the inverse model also reconstructs the internal concentration field with reasonable accuracy. Over the selected evaluation times, the reconstructed profiles achieve a mean relL2 error of 3.77×10^{-2} . This indicates that the inverse SDD-PINN provides a proof of concept for parameter recovery and state reconstruction in a model-consistent synthetic setting, with the reconstructed spatial state remaining consistent with the governing PDE.

Overall, this synthetic example demonstrates that SDD-PINN can serve not only as a forward solver but also as a practical inverse framework for shrinking-domain droplet-drying problems. The present case is intentionally limited to a model-consistent setting with synthetic observations and known initial profile. Nevertheless, it provides a first proof of concept that sparse noisy measurements can be combined with the physics-constrained SDD-PINN formulation to recover unknown

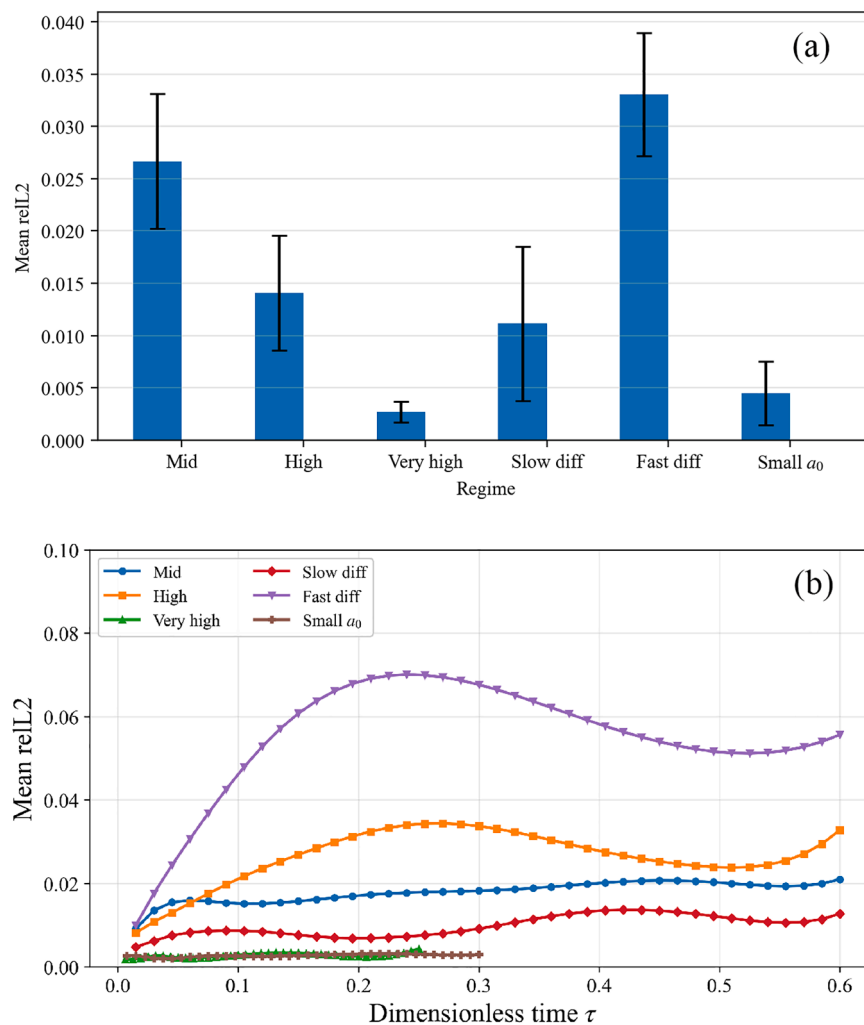


Fig. 12. Accuracy of the trained SDD-PINN across different drying regimes relative to the CN reference: mean relL2 error (a) across seeds; (b) versus dimensionless time τ .

drying parameters and reconstruct the internal droplet state. Extension to additional unknowns, such as interfacial transfer parameters and more realistic experimental observables, is a natural next step.

5. Conclusion

A compact, reproducible PINN recipe tailored to shrinking-domain diffusion in a spherical single droplet setting with a Robin BC was developed. The key ingredients were hard enforcement of the IC, the symmetry-consistent input feature, ξ^2 , a log-time reparameterization (θ), simple but structured collocation, and a combined Adam + L-BFGS optimizer schedule. These substantially mitigated key failure modes of the vanilla PINN (IC sticking, center singularity artifacts, late-time drift). On the reference regime, the final configuration achieved a mean relL2 error of 0.021 ± 0.010 versus a CN benchmark. The model was also tested across additional drying regimes spanning shrinkage intensity, diffusivity, and initial size, for which the regime-averaged mean relL2 remained between 2.69×10^{-3} and 3.30×10^{-2} . This SDD-PINN recipe is simple to implement within this problem class: each tweak is easy to add, needs no specialized architectures or elaborate loss-weighting schemes, and the selected progressive path showed steady gains as the remedies were introduced progressively.

This study is deliberately scoped to diffusion-dominant transport with geometric shrinkage. Extending the approach to coupled heat and mass transfer, crust formation, and more practical inverse identification

using experimental observables and additional unknown parameters are natural next steps. Taken together, the results suggest that carefully chosen, general-purpose PINN practices can deliver stable, accurate solutions for this class of shrinking-domain diffusion problems at modest implementation cost, providing a strong baseline for more complex single-droplet drying models and related multiphysics systems.

CRediT authorship contribution statement

Narjes Malekjani: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Andreas Bück:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization. **Abdolreza Kharaghani:** Writing – review & editing, Supervision. **Evangelos Tsotsas:** Writing – review & editing, Supervision, Project administration, Conceptualization.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Narjes Malekjani reports financial support was provided by Alexander von Humboldt Foundation. Andreas Bück reports financial support was provided by German Research Foundation. Reports a

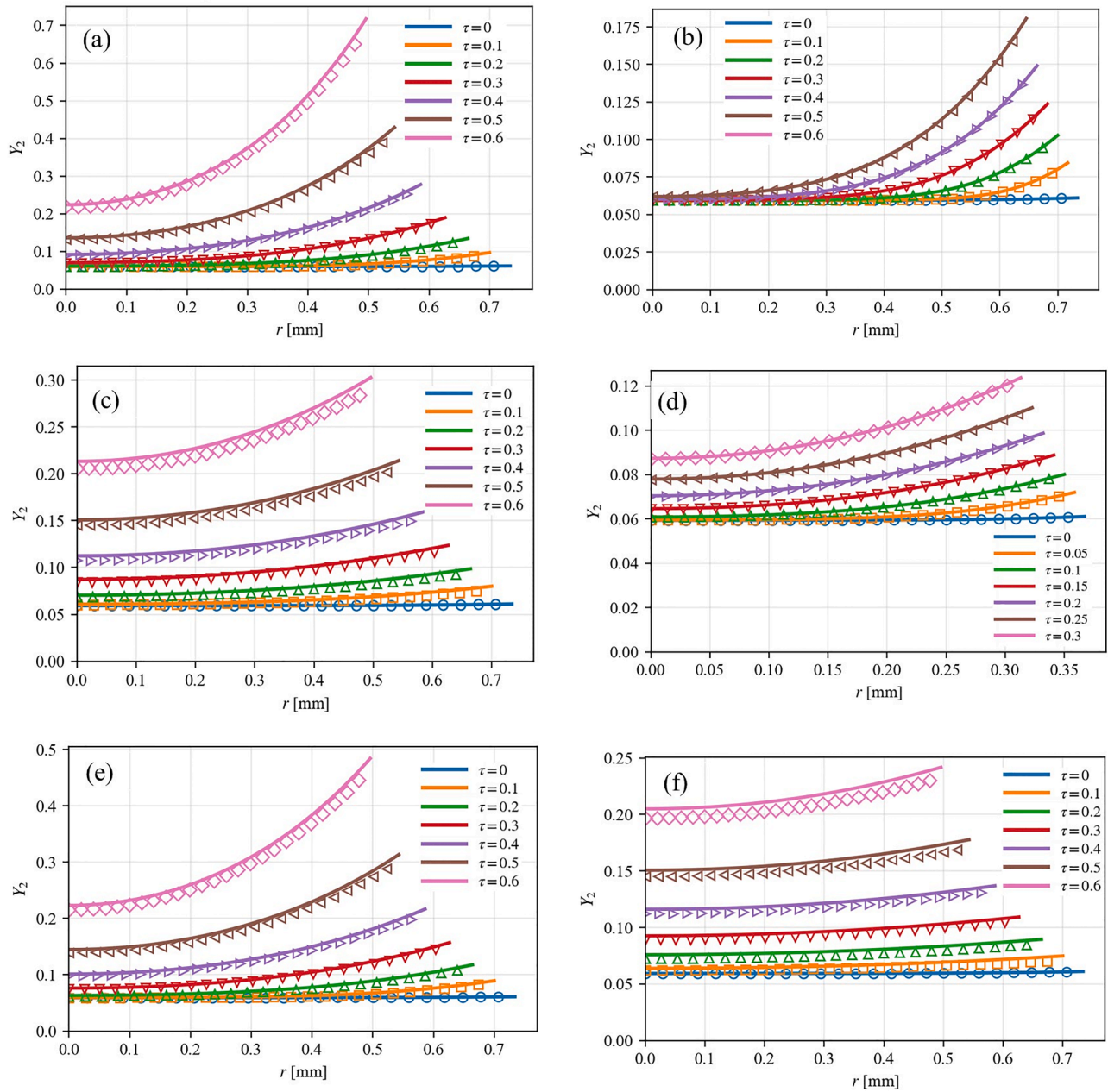


Fig. 13. Performance of SDD-PINN in different drying regimes (a) high, (b) very high, (c) mid, (d) small a_0 , (e) slow diff, (f) fast diff compared to CN (solid lines).

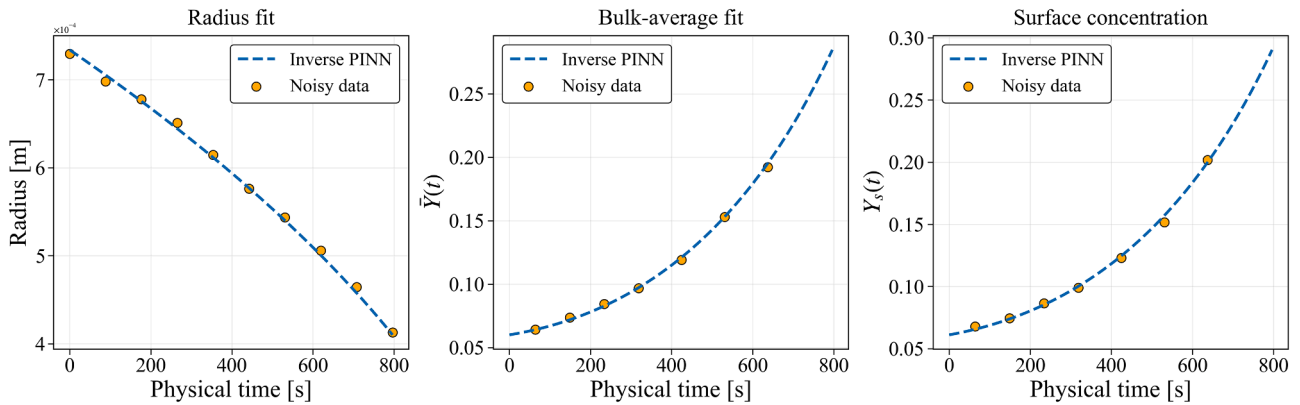


Fig. 14. Synthetic inverse-observation fits for the reference drying regime.

relationship with that includes: Has patent pending to. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Transformation of moving to fixed computational domain

In the spherical coordinates, we have the diffusion equation as:

$$\frac{\partial Y_i}{\partial t} = D \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial Y_i}{\partial r} \right), \quad 0 \leq r \leq a(t). \quad (\text{A.1})$$

We map the physical radius $r \in [0, a(t)]$ to a fixed domain $\xi = \frac{r}{a(t)} \in [0, 1]$ so $r = a(t) \xi$. At fixed t , $a(t)$ is constant hence:

$$\left. \frac{\partial}{\partial r} \right|_t = \frac{1}{a(t)} \left. \frac{\partial}{\partial \xi} \right|_t, \quad \left. \frac{\partial^2}{\partial r^2} \right|_t = \frac{1}{a(t)^2} \left. \frac{\partial^2}{\partial \xi^2} \right|_t. \quad (\text{A.2})$$

For the spherical operator:

$$\left. \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial Y_i}{\partial r} \right) \right|_t = \frac{1}{a^2 \xi^2} \left. \frac{\partial}{\partial \xi} \right|_t \left(\xi^2 \frac{\partial Y_i}{\partial \xi} \right). \quad (\text{A.3})$$

Because $r = \xi a(t)$, holding r fixed gives $dr = a d\xi + \xi da = 0$, so:

$$\left. \frac{d\xi}{dt} \right|_r = -\frac{\xi}{a} \frac{da}{dt} = -\frac{\dot{a}}{a} \xi. \quad (\text{A.4})$$

Then:

$$\left. \frac{\partial Y_i}{\partial t} \right|_r = \left. \frac{\partial Y_i}{\partial t} \right|_\xi + \left. \frac{\partial Y_i}{\partial \xi} \right|_t \left. \frac{d\xi}{dt} \right|_r = \left. \frac{\partial Y_i}{\partial t} \right|_\xi - \frac{\dot{a}}{a} \xi \left. \frac{\partial Y_i}{\partial \xi} \right|_t. \quad (\text{A.5})$$

For transforming Eq. (A.1), we use Eqs. (A.3) and (A.5) to obtain:

$$\left. \frac{\partial Y_i}{\partial t} \right|_\xi - \frac{\dot{a}}{a} \xi \left. \frac{\partial Y_i}{\partial \xi} \right|_t = \frac{D}{a^2 \xi^2} \left. \frac{\partial}{\partial \xi} \right|_t \left(\xi^2 \frac{\partial Y_i}{\partial \xi} \right). \quad (\text{A.6})$$

Multiplying by a^2 and using $a\dot{a} = \frac{1}{2} \frac{d(a^2)}{dt}$ gives the fixed domain equation:

$$a^2 \frac{\partial Y_i}{\partial t} = \frac{1}{2} \frac{d(a^2)}{dt} \xi \left. \frac{\partial Y_i}{\partial \xi} \right|_t + \frac{D}{\xi^2} \left. \frac{\partial}{\partial \xi} \right|_t \left(\xi^2 \frac{\partial Y_i}{\partial \xi} \right). \quad (\text{A.7})$$

Appendix B. Time-scaling of process model

To convert Eq. (15) to Eq. (25) using the definition of new time $\theta = -\ln(1 + \alpha\tau)$ we used the chain rule between τ and θ :

$$\theta = -\ln(1 + \alpha\tau) \rightarrow \frac{d\theta}{d\tau} = -\frac{\alpha}{1 + \alpha\tau}, \quad (\text{B.1})$$

$$\frac{\partial}{\partial \tau} = \frac{d\theta}{d\tau} \frac{\partial}{\partial \theta} = -\frac{\alpha}{1 + \alpha\tau} \frac{\partial}{\partial \theta} \leftrightarrow \frac{\partial}{\partial \theta} = \frac{d\tau}{d\theta} \frac{\partial}{\partial \tau} = -\frac{1 + \alpha\tau}{\alpha} \frac{\partial}{\partial \tau}, \quad (\text{B.2})$$

$$\frac{\partial Y_i}{\partial \theta} = \frac{d\tau}{d\theta} \frac{\partial Y_i}{\partial \tau} = -\frac{1 + \alpha\tau}{\alpha} \frac{\partial Y_i}{\partial \tau}. \quad (\text{B.3})$$

Eq. (15) becomes:

$$\frac{\partial Y_i}{\partial \theta} = -\frac{G}{\alpha} \left[\frac{\partial^2 Y_i}{\partial \xi^2} + \left(\frac{2}{\xi} - 2\delta\xi \right) \frac{\partial Y_i}{\partial \xi} \right] \quad (\text{B.4})$$

Appendix C. Supplementary training diagnostics along the progressive enhancement path

This appendix provides the detailed diagnostic plots referenced in the main text, illustrating the main training pathologies in PINNs for the shrinking-droplet drying problem and how these pathologies evolve along the progressive enhancement path. For readers unfamiliar with deep-learning training dynamics, we provide accessible explanations below. All curves report the median with interquartile range (IQR) across training

seeds. Steps 0–3 correspond to the progressive enhancement path used in the main text. In Steps 2 and 3, the dashed vertical line marks the transition from Adam to L-BFGS refinement.

To make these diagnostics precise, let w denote the trainable network parameters. The composite loss is written as

$$Loss_{total}(w) = \omega_{PDE} Loss_{PDE} + \omega_{center} Loss_{center} + \omega_{surface} Loss_{surface} + \omega_{IC} Loss_{IC}, \quad (C1)$$

where the individual terms are the mean-squared residuals of the PDE, the IC, the center BC, and the surface BC, respectively. For configurations with hard IC enforcement, IC loss is identically zero and is omitted from the optimized objective. In the gradient diagnostics below, BC denotes the combined boundary-related contribution used in the logged training diagnostics.

C.1. Gradient norms along the progressive enhancement path

Gradient norms quantify how strongly each loss component drives parameter updates. For each loss component $Loss_i$, we report the Euclidean norm of its parameter gradient

$$\|\nabla_w Loss_i\|_2 = \sqrt{\sum_p \left(\frac{\partial Loss_i}{\partial w_p} \right)^2}, \quad (C2)$$

evaluated on a fixed diagnostic batch during optimization. Large disparities among these norms indicate that one loss component dominates the update direction and therefore controls training disproportionately, even if it is not the only large residual term. In this sense, gradient norms provide a direct diagnostic of optimization imbalance.

In Fig. C1:

- Step 0 (baseline). The PDE gradient norm is orders of magnitude larger than the IC and BC gradient norms. This indicates that the optimizer is driven mainly by the stiff interior PDE term, while the constraint-related terms exert comparatively weak influence on the parameter updates.
- Step 1 (+ hard IC). Hard IC removes the competing IC-loss gradient entirely from the optimization problem. This simplifies the early optimization dynamics, but the PDE gradient still remains strongly dominant over the boundary-related gradients, so the main imbalance persists. Accordingly, Step 1 improves accuracy by removing unnecessary competition with the IC, but does not yet resolve the dominant interior training difficulty.
- Step 2 (+ ξ^2 and Adam + L-BFGS). The gap between PDE-driven and boundary-driven gradients is reduced relative to Step 1. This indicates that the parameter updates are now influenced by a more balanced combination of interior and boundary information. This step should be interpreted as a coupled improvement, not as an isolated ξ^2 -only ablation: the ξ^2 mapping improves the conditioning of the spatial representation, while the optimizer upgrade provides a more effective late-stage refinement once Adam has reached a reasonable parameter region.
- Step 3 (+ θ -time reparameterization). The gradient histories remain broadly comparable in scale to Step 2, but the final predictive accuracy improves substantially. This indicates that the Step 3 advantage is not expressed most clearly as a dramatic collapse of the gradient norms alone. Instead, its main benefit is seen more directly in the suppression of late-time error growth and in the stronger final-stage reduction of the residual histories reported in Fig. C2.

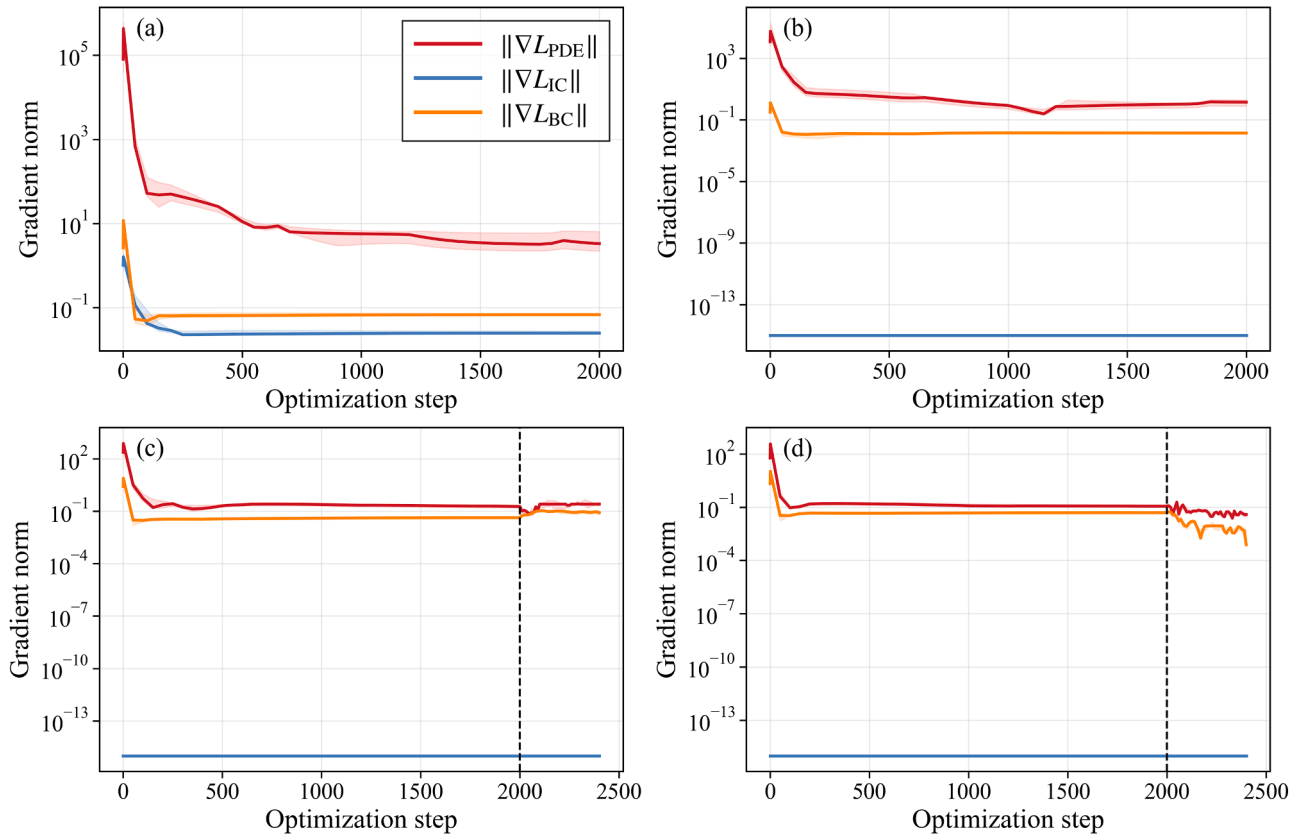


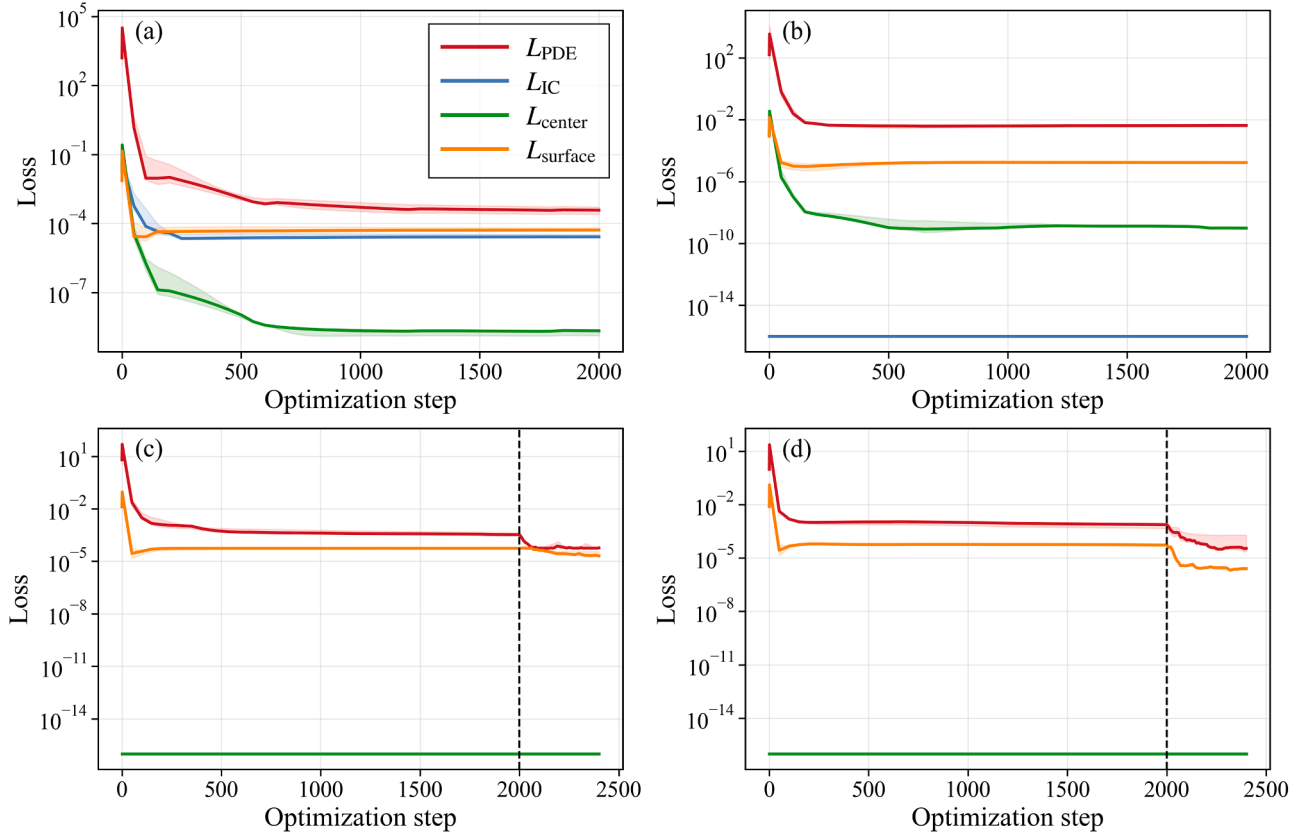
Fig. C1. Gradient-norm histories along the progressive enhancement path (a) step 0, (b) step 1, (c) step 2, (d) step 3.

C.2. Residual imbalance over optimization

The residual-imbalance plots decompose the total loss into its components (PDE interior residual, IC residual, and center/surface boundary residuals) and track their magnitudes over optimization steps. Residual imbalance refers to the situation in which some components decrease rapidly whereas others remain large or plateau. In PINNs, this typically indicates that some constraints are easier to satisfy than others, or that different terms have different scales and effective stiffness.

In Fig. C2:

- Step 0 (baseline). The PDE residual remains dominant throughout training and decays much more slowly than the IC and BC terms. This shows that the baseline model can satisfy the easier constraint terms more readily than the interior transport dynamics, which is consistent with the late-time prediction errors reported in the main text.
- Step 1 (+ hard IC). Enforcing the IC analytically removes the need to learn it through a competing soft penalty. The early-stage competition is therefore reduced, but the PDE residual still remains substantially larger than the boundary-related terms. This confirms that hard IC alone does not eliminate the dominant interior training difficulty.
- Step 2 (+ ξ^2 and Adam + L-BFGS). The PDE residual becomes markedly smaller than in Step 1, and the imbalance between the dominant interior residual and the easier boundary terms is reduced. After the optimizer switch, the remaining residuals decrease further. This behavior is consistent with the interpretation adopted in the main text: Step 2 represents a combined improvement in which ξ^2 improves the conditioning of the spatial representation and the optimizer upgrade then enables more effective late-stage minimization.
- Step 3 (+ θ -time reparameterization). The residual curves reach the lowest final PDE and surface-loss levels overall, and the dominant imbalance present in the earlier steps is reduced further. This is the clearest evidence for the Step 3 gain: once the late-time amplification intrinsic to the original τ -formulation is mitigated, the residuals can be reduced more effectively across the full drying interval.



• C2Residual-imbalance histories along the progressive enhancement path (a) step 0, (b) step 1, (c) step 2, (d) step 3.

C.3. Operator-term diagnostics along the progressive enhancement path

These plots monitor the relative magnitudes of the main transformed PDE terms at the collocation points. In the baseline τ -formulation, the governing operator can be written schematically as a balance between the time derivative and a scaled spatial operator,

$$Y_\tau = c(\tau) \left[Y_{\xi\xi} + \left(\frac{2}{\xi} - 2s\xi \right) Y_\xi \right], \quad (C3)$$

where

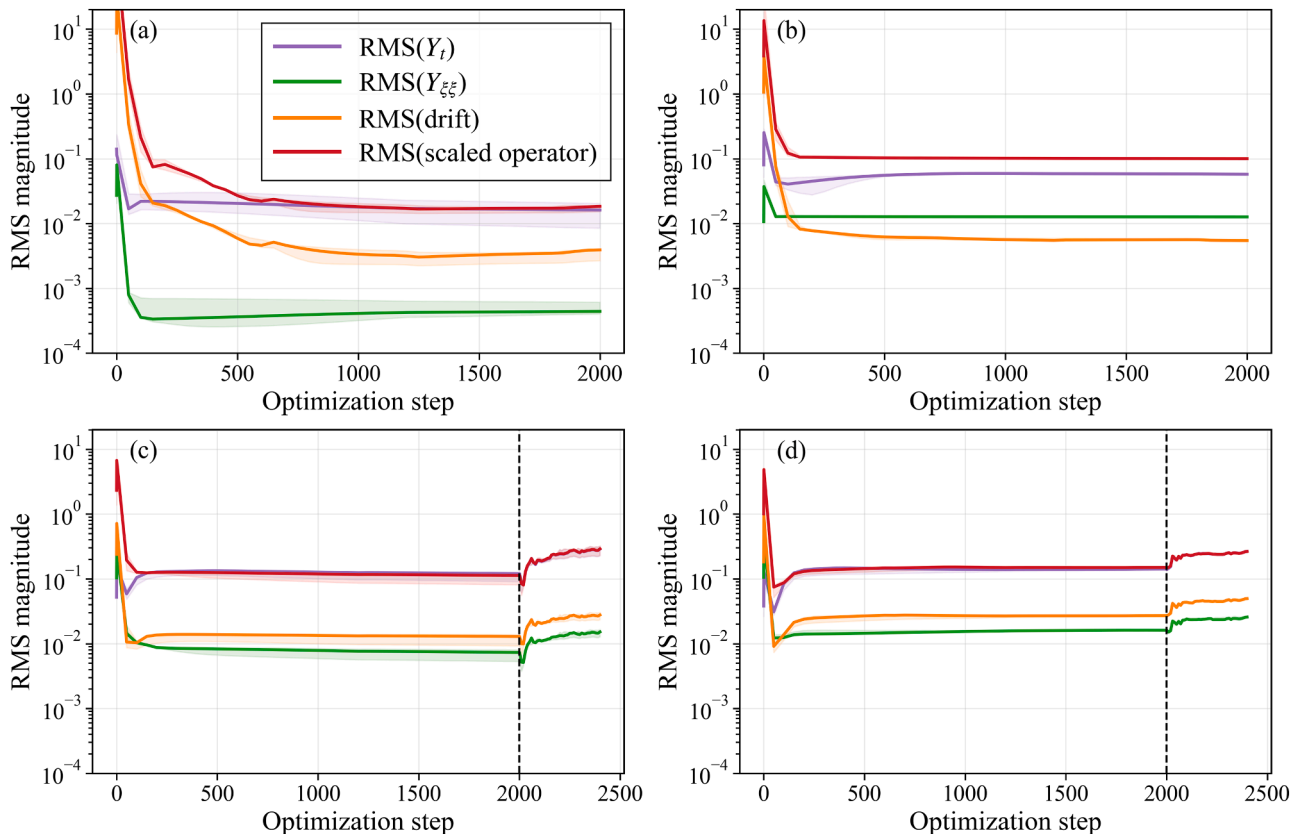
$$c(\tau) = \frac{G}{1 + \alpha\tau}. \quad (C4)$$

We therefore track the root-mean-square (RMS) magnitudes of the following quantities during training: $\text{RMS}(Y_\tau)$, $\text{RMS}(Y_{\xi\xi})$, $\text{RMS}\left(\left(\frac{2}{\xi} - 2s\xi\right)Y_\xi\right)$, $\text{RMS}\left(c(\tau)\left[Y_{\xi\xi} + \left(\frac{2}{\xi} - 2s\xi\right)Y_\xi\right]\right)$, which quantify scale separation among operator terms and therefore the effective stiffness seen during optimization.

In Fig. C3:

- Step 0 (baseline). The scaled spatial operator remains comparatively large during training, indicating persistent operator amplification and effective stiffness growth. This is consistent with the poor late-time predictions in the baseline and with the observed PDE-gradient dominance.
- Step 1 (+ hard IC). Hard IC improves constraint handling, but does not materially change the structure of the PDE operator itself. Accordingly, the scaled operator remains large, showing that the main stiffness source has not yet been removed.
- Step 2 (+ ξ^2 and Adam + L-BFGS). The relative magnitudes of the diffusion and drift-related terms become better behaved than in the earlier steps, consistent with improved spatial conditioning and more effective optimizer refinement. However, because the original τ -formulation is still used, the time-dependent amplification of the scaled spatial operator is not removed at this stage. Accordingly, the operator-term diagnostics are consistent with the interpretation that Step 2 improves the optimization landscape and residual reduction, but does not yet eliminate the late-time amplification intrinsic to the original formulation.
- Step 3 (+0-time reparameterization). The θ -transformation removes the growing prefactor present in the original τ -formulation and converts the problem to a form with much more uniform operator scaling over the drying interval at the equation level. In the diagnostics, this effect is reflected more clearly in the overall training behavior than in any single operator trace alone. Relative to Step 2, Step 3 shows much stronger suppression of

late-time error growth in the main text and stronger final-stage reduction of the PDE and surface residuals in Fig. C2. The operator histories here are therefore best interpreted as supporting evidence: they are consistent with the view that the θ -transformation mitigates the late-time amplification present in the original τ -formulation, but the clearest evidence for the Step 3 advantage appears in the error-propagation and residual-reduction plots rather than in the operator traces alone.



. C3Operator-term diagnostics along the progressive enhancement path (a) step 0, (b) step 1, (c) step 2, (d) step 3.

Appendix D. Detailed inverse example plots

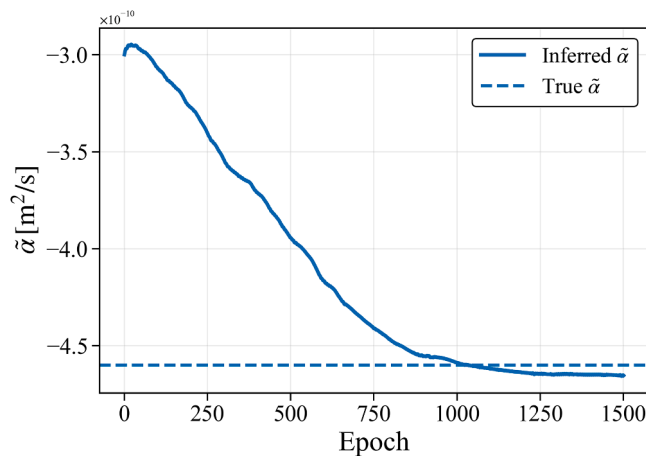


Fig. D1. Convergence history of the recovered shrinkage parameter $\tilde{\alpha}$ during inverse training, showing approach to the true value.

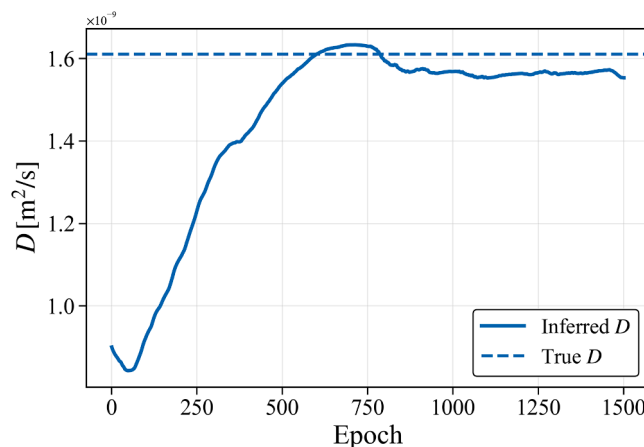


Fig. D2. Convergence history of the recovered diffusion coefficient D during inverse training, showing stabilization near the true value.

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